

Robotic Arm Gripping Anomaly Solutions

If after completing the "Calibrate Robotic Arm" and "Calibrate Offset" courses, the robotic arm still cannot **grip tightly**, you can refer to the following tutorial to update the expansion board firmware.

Important Notes

Note: If the robotic arm cannot grip accurately (rather than tightly), please first check whether the suction cup is firmly attached.

If the suction cup is not firmly attached, it will cause a height difference between the robotic arm and the desktop, which will directly lead to gripping position offset, resulting in inaccurate gripping. **In this case, firmware upgrade is not required.**

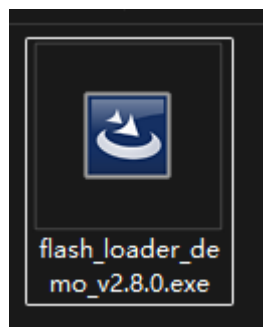
Hardware Preparation

- EVV01 Robotic Arm Servo Control Board
- MicroUSB Data Cable

Software Tools

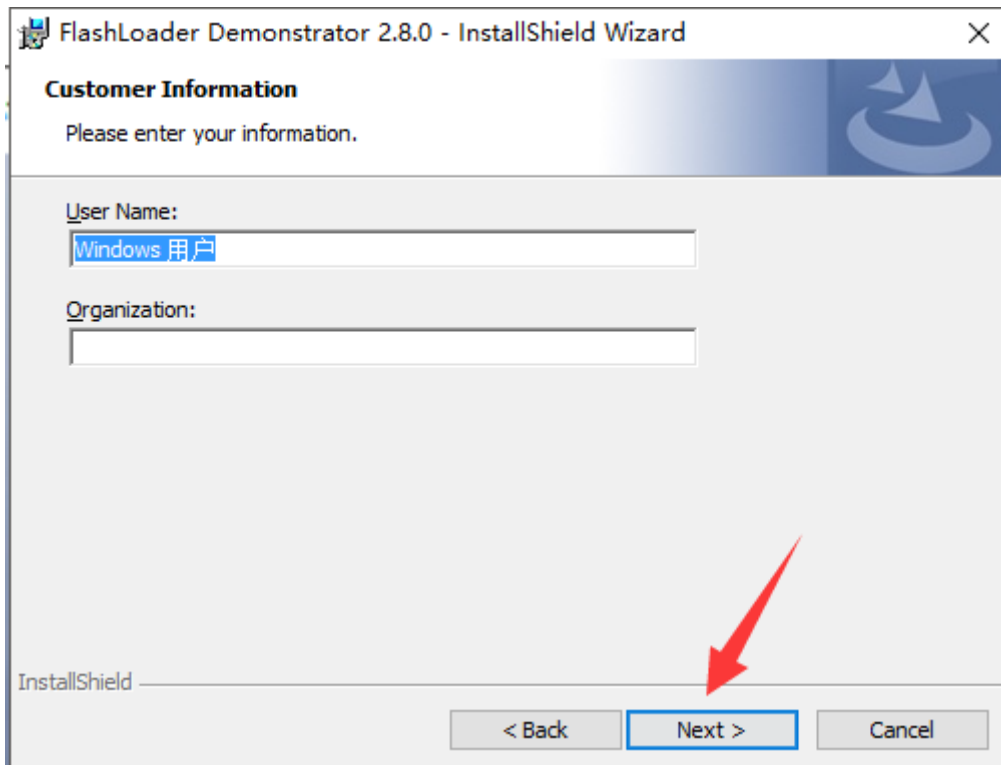
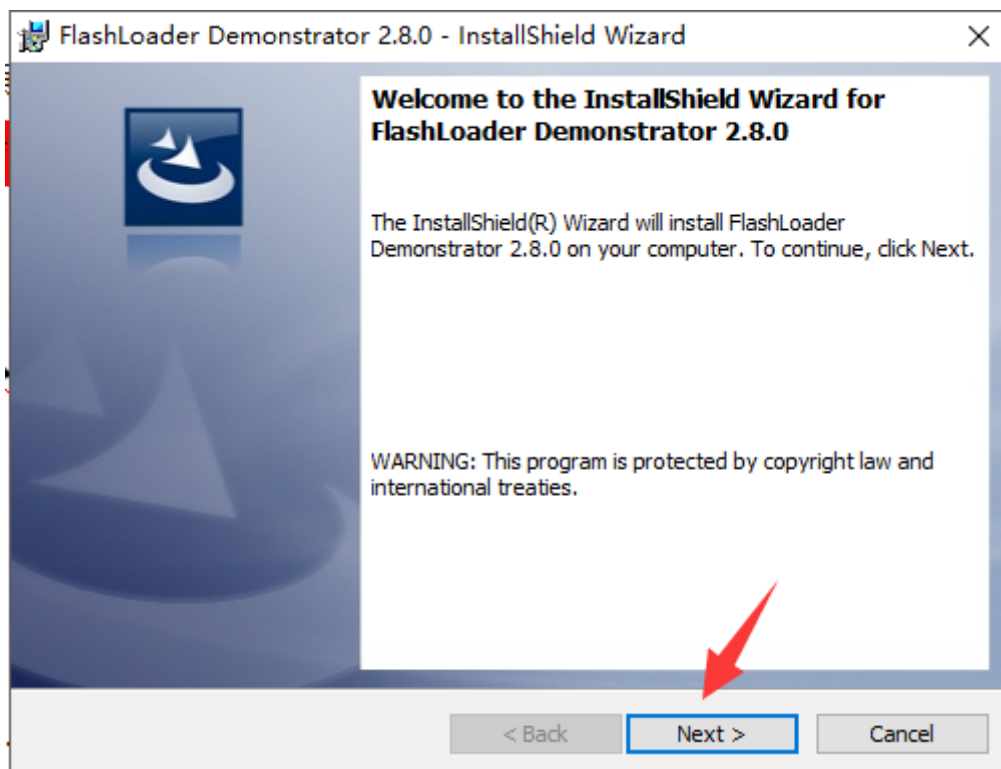
The upgrade requires the following software tools:

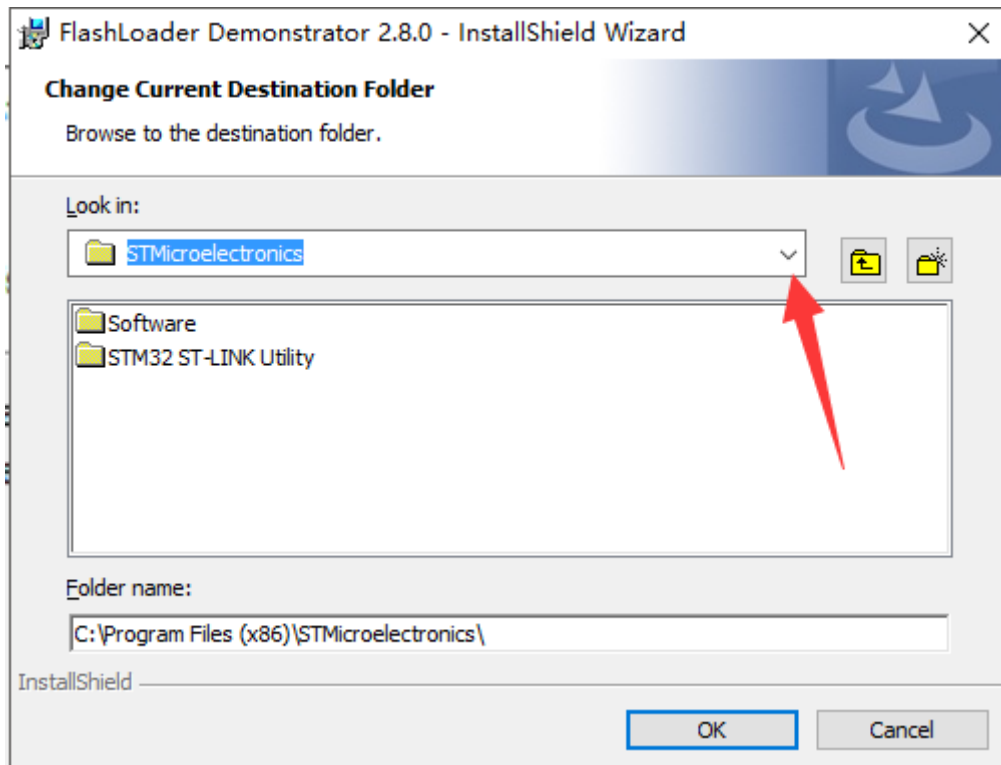
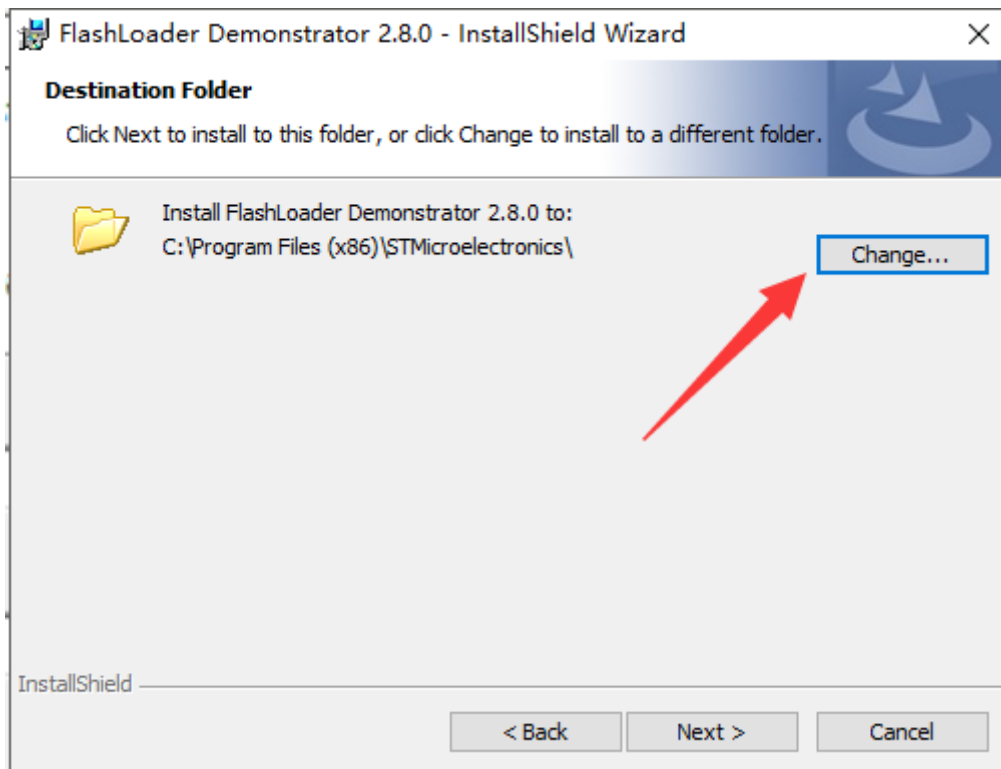
Tool	Description
New Firmware	<code>arm_slave_vxxxx.hex</code>
Demonstrator GUI	Needs to be installed via <code>flash_loader_demo_v2.8.0.exe</code> from the accompanying materials

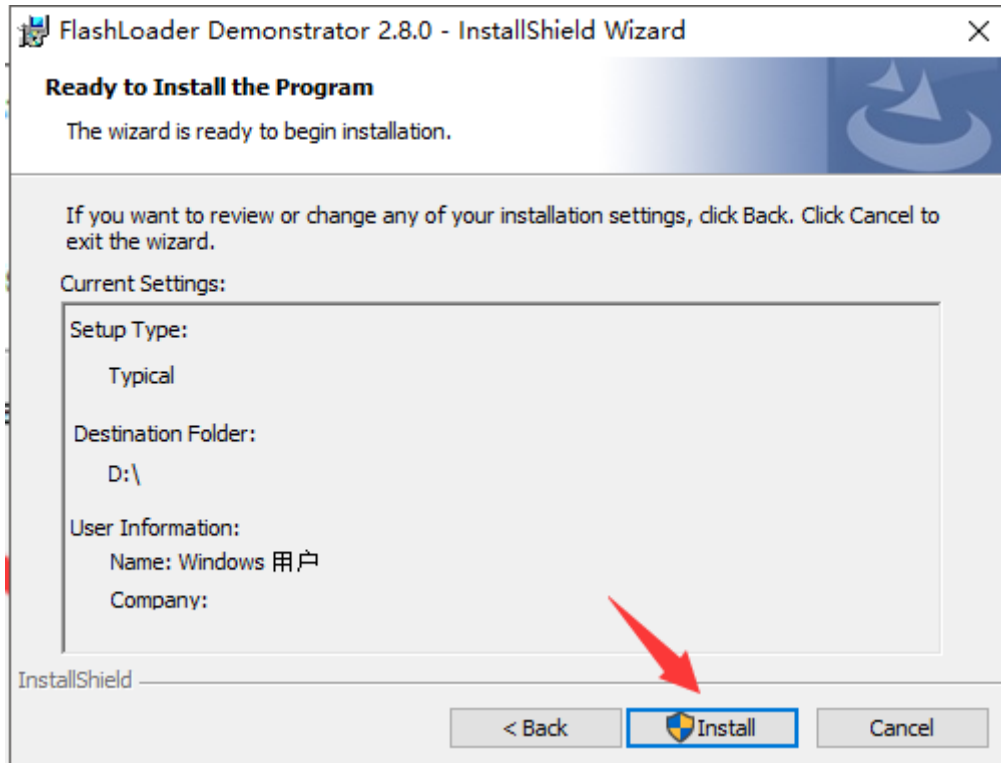
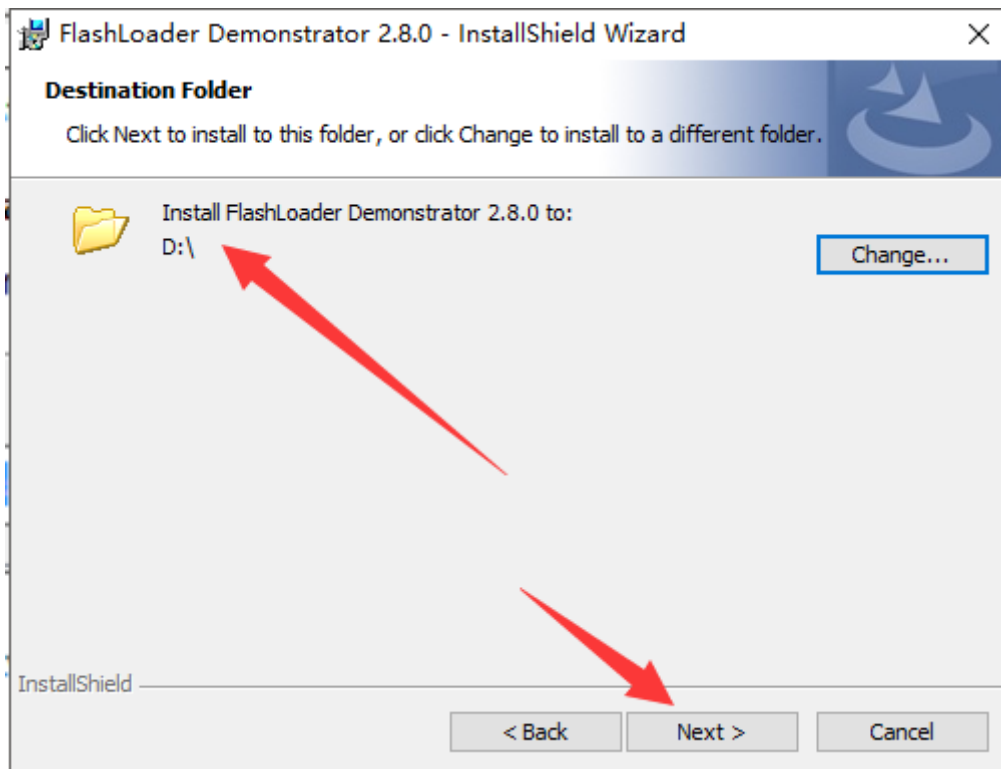


Installation Steps

The following are screenshots of the software installation process, follow them in order.

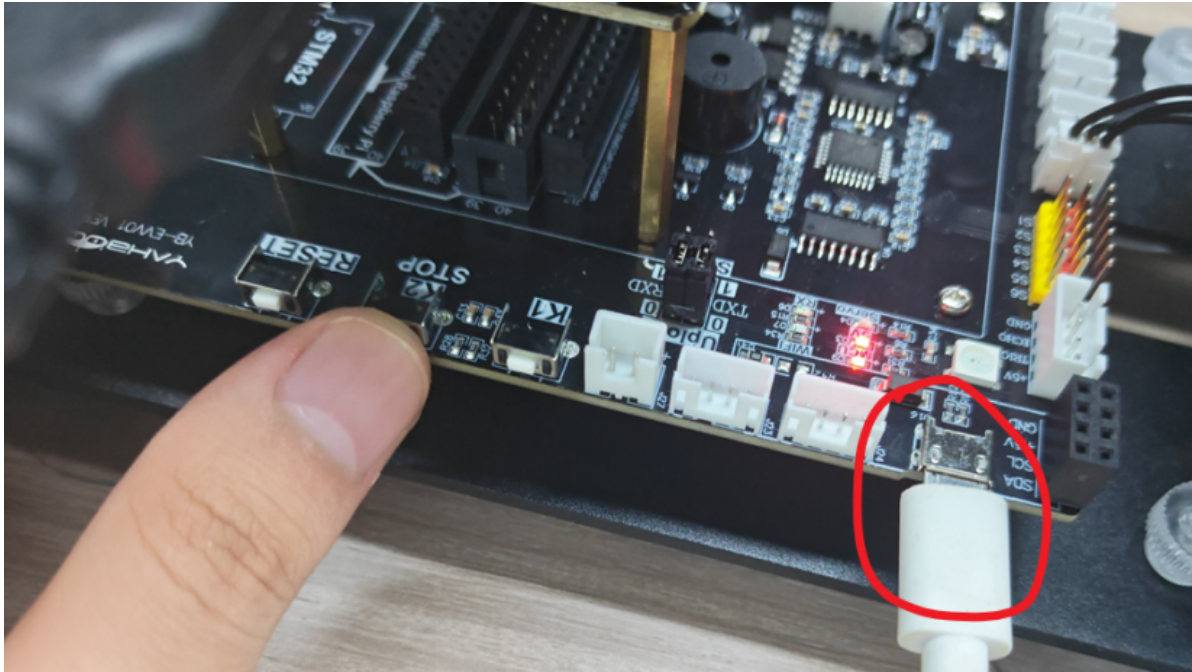






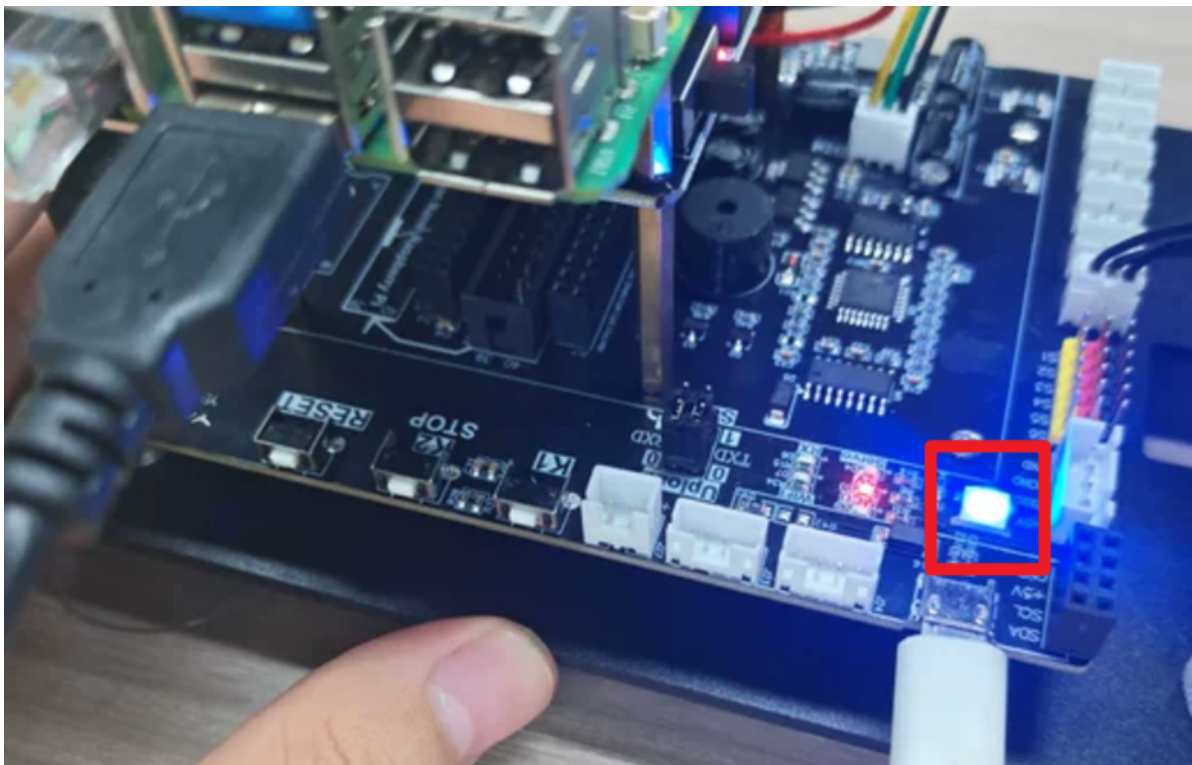
2. Connect to Computer

Use a **MicroUSB cable** to connect the computer to the servo expansion board.



3. Enter Programming Mode

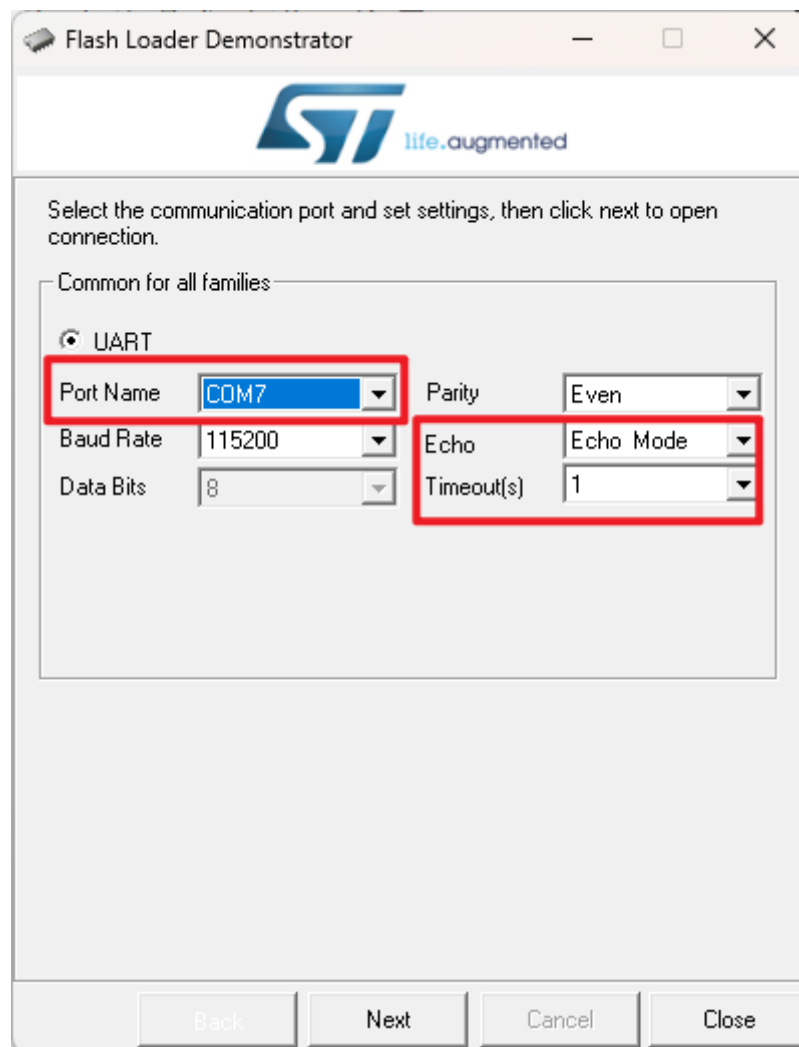
Press and hold the K2 (STOP) button for 15 seconds. The RGB LED on the mainboard will enter a running light mode, indicating successful entry into programming mode.



4. Open Demonstrator GUI and Configure Serial Port

Open the **Demonstrator GUI** software installed earlier:

- **Serial Port:** Select the serial port corresponding to MicroUSB
- **Other Options:** Select as shown in the diagram



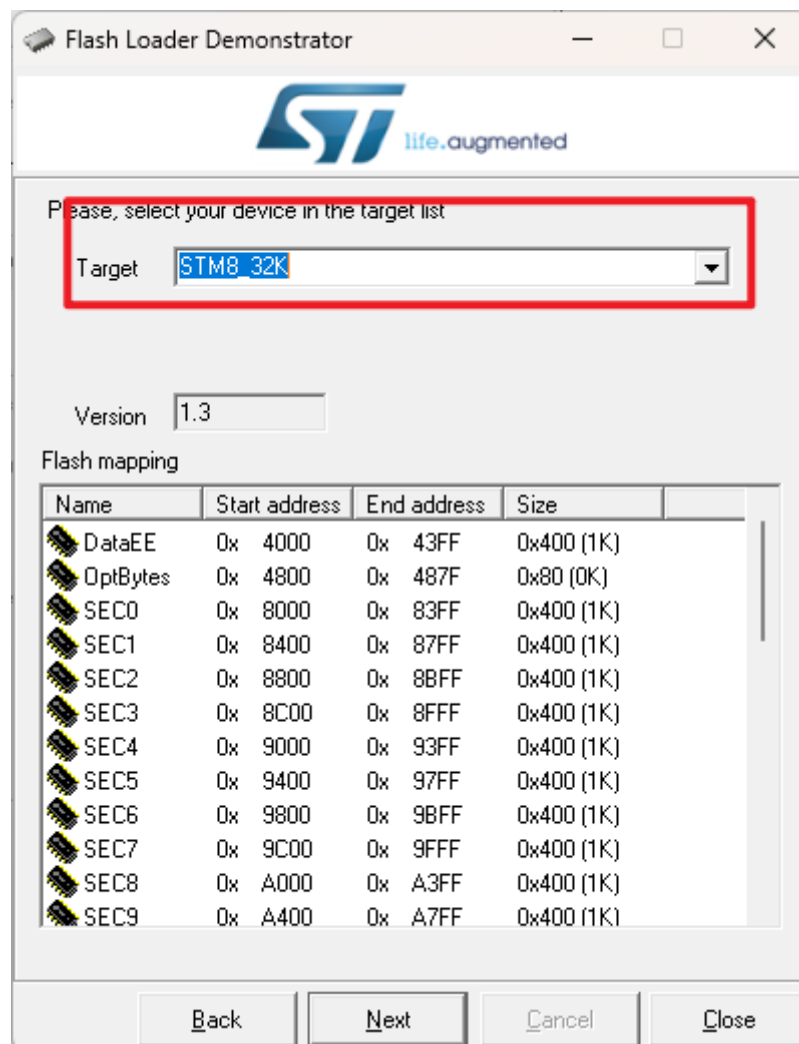
5. Reset and Proceed to Next

After selection, **press the RESET button on the mainboard, then immediately click the Next button in the software** (the interval must not exceed 1 second).



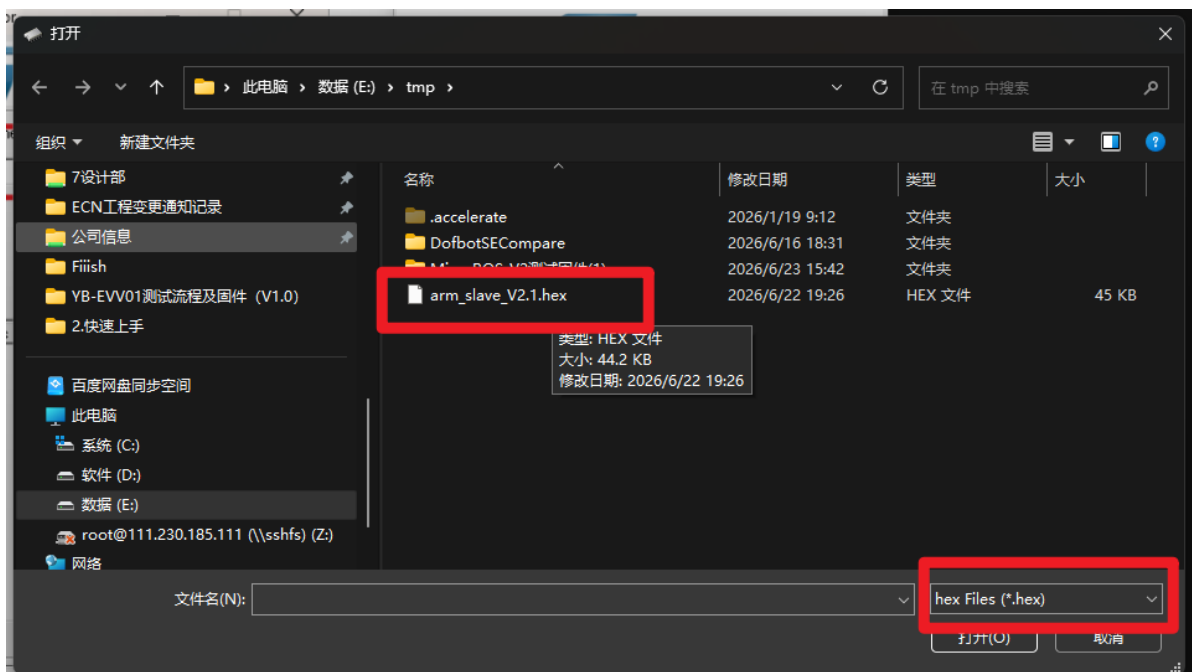
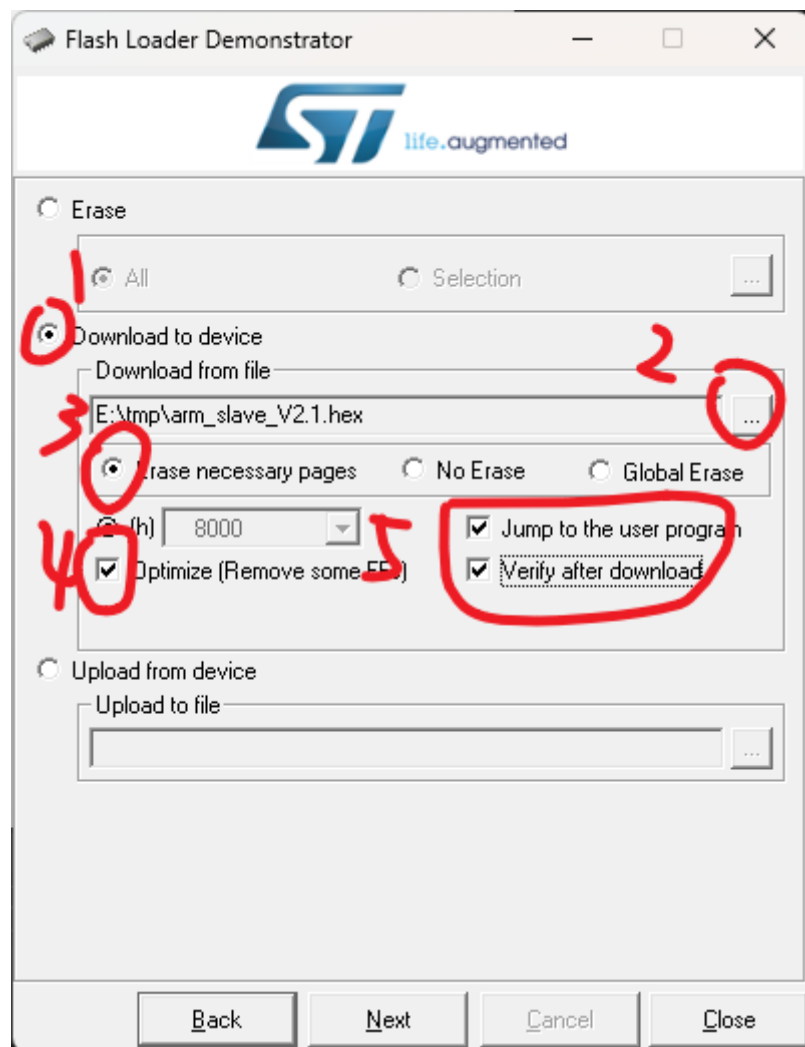
6. Select Target Chip

Select **STM8_32K** as the target, then click Next.



7. Select Firmware to Program

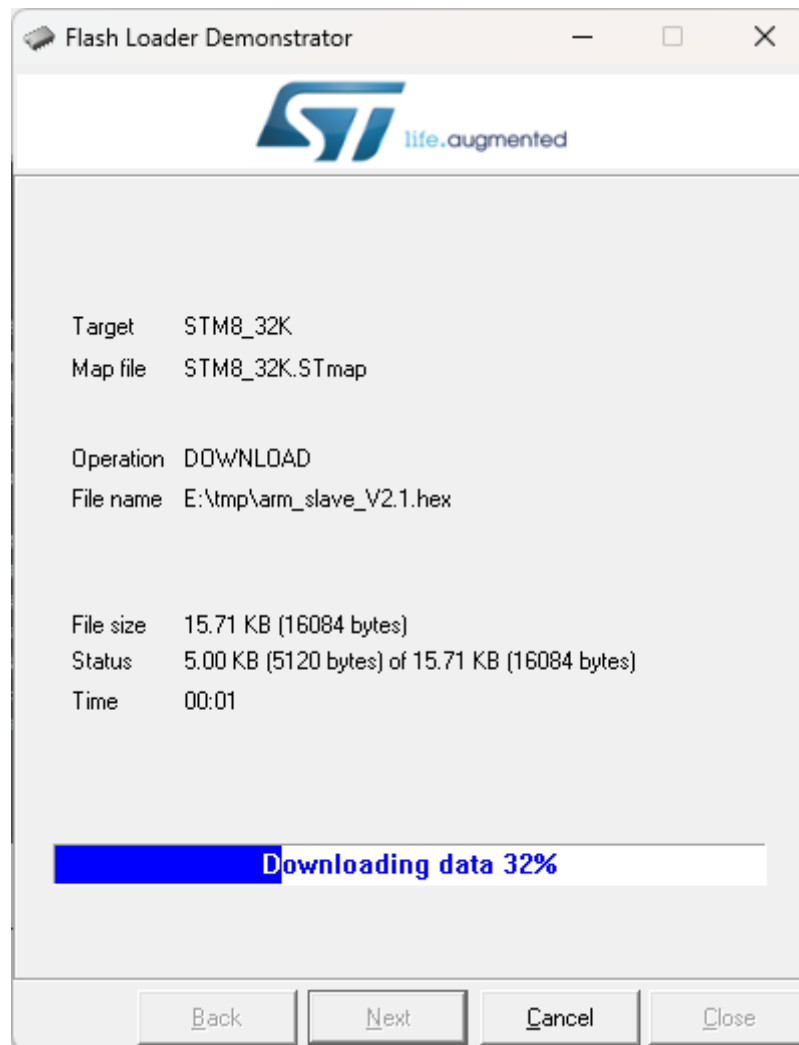
Select the firmware file to program (arm_slave_vxxxx.hex).



After selection, click Next to enter the programming interface.

8. Wait for Programming to Complete

Wait for the progress bar to turn green, indicating that programming is complete.



Note: If clicking Next does not respond at this step, please ensure that no other main controller (such as STM32, etc.) is connected to the servo expansion board.

After programming is complete, press and hold the K2 button on the servo expansion board for 15 seconds again to exit programming mode (the rainbow running light turns off), and remember to **return the two jumper caps to their original positions, placing them back on the Servo side.**